Test Bank

CH43

MCQ:

1) What is the primary function of the right ventricle in the mammalian heart?

A) To pump oxygen-rich blood to the body
B) To pump oxygen-poor blood to the lungs
C) To receive oxygen-rich blood from the lungs

Answer: B

2) Which blood vessel carries oxygen-rich blood from the lungs to the heart?

A) Pulmonary artery
B) Aorta
C) Pulmonary vein

Answer: C

3) What causes the 'lub' sound in a heartbeat?

A) Recoil of blood against the semilunar valves
B) Recoil of blood against the atrioventricular (AV) valves
C) Contraction of the ventricles

Answer: B

4) What is the main function of erythrocytes?

A) To transport oxygen
B) To fight infections

C) To clot blood

Answer: A

5) What is the primary function of capillaries in the circulatory system?

A) To facilitate the exchange of materials between blood and tissues
B) To prevent backflow of blood
C) To transport blood at high pressure

Answer: A

6) What is the primary cause of a heart attack?

A) Blockage of coronary arteries
B) High blood pressureLow cholesterol levels
C)

Answer: A

7) What is the Bohr shift in relation to hemoglobin?

A) An increase in hemoglobin's affinity for oxygen at lower pH
B) A decrease in hemoglobin's affinity for oxygen at lower pH
C) A change in hemoglobin's structure at high altitudes

Answer: B

8) What is the function of surfactants in the lungs?

A) To coat the alveoli and reduce surface tension
B) To transport oxygen in the blood
C) To increase the surface area for gas exchange

Answer: A

9) What is the primary method of CO_2 transport in the blood?

A) Dissolve in plasma
B) Bound to hemoglobin
C) As bicarbonate ions

Answer: C

10) What is the primary function of the left ventricle in the mammalian heart?

A) To pump oxygen-poor blood to the lungs
B) To receive oxygen-poor blood from the body
C) To pump oxygen-rich blood to the body

Answer: C

11) What is the role of the atrioventricular (AV) valves in the heart?

A) To separate the atria from the ventricles
B) To control blood flow to the lungs
C) To regulate the heart's rhythm

Answer: A

12) What causes the 'dup' sound in a heartbeat?

A) Recoil of blood against the atrioventricular (AV) valves
B) Contraction of the atria
C) Recoil of blood against the semilunar valves

Answer: C

13) What is the main function of leukocytes?

A) To transport oxygen
B) To fight infections

C) To carry nutrients

Answer: B

14) What is the primary function of platelets in the blood?

A) To transport carbon dioxide
B) To fight infections
C) To aid in blood clotting

Answer: C

15) What is the primary cause of a stroke?

A) Blockage or rupture of arteries in the brain
B) High cholesterol levels
C) Low blood pressure

Answer: A

16) What is the function of the pulmonary artery?

A) To carry oxygen-rich blood to the lungs
B) To carry oxygen-poor blood to the lungs
C) To carry oxygen-rich blood to the heart

Answer: B

17) What is the primary role of hemoglobin in red blood cells?

A) To transport carbon dioxide
B) To transport oxygen
C) To fight infections

Answer: B

18) What is the primary function of the aorta in the circulatory system?

A) To carry oxygen-poor blood to the lungs
B) To carry oxygen-rich blood from the heart to the body
C) To carry oxygen-poor blood from the body to the heart

Answer: B

19) Which part of the heart receives oxygen-poor blood from the body?

A) Right atrium
B) Left ventricle
C) Left atrium

Answer: A

20) What is the main function of the pulmonary veins?

A) To carry oxygen-poor blood to the lungs
B) To carry oxygen-rich blood from the lungs to the heart
C) To carry oxygen-rich blood to the body

Answer: B

21) What is the role of the semilunar valves in the heart?

A) To separate the atria from the ventricles
B) To prevent backflow of blood into the ventricles
C) To regulate the heart's rhythm

Answer: B

22) What is the primary function of the superior vena cava?

A) To carry oxygen-rich blood from the lungs to the heart

B) To carry oxygen-rich blood to the brain
C) To carry oxygen-poor blood from the upper body to the heart

Answer: C

23) What is the main function of the coronary arteries?

A) To supply blood to the heart muscle
B) To carry blood to the brain
C) To transport blood to the lungs

Answer: A

24) What is the primary role of the diaphragm in respiration?

A) To filter air entering the lungs
B) To contract and expand the chest cavity for breathing
C) To exchange gases in the alveoli

Answer: B

25) What is the function of the capillary beds in the lungs?

A) To transport nutrients to the body
B) To exchange oxygen and carbon dioxide with the blood
C) To filter impurities from the blood

Answer: B

26) What is the primary function of the lymph nodes?

A) To produce red blood cells
B) To fill lymph and fight infections
C) To transport oxygen to tissues

Answer: B

CH47

MCQ:

1) What is the first line of defense in innate immunity for insects?

A) Skin
B) Exoskeleton made of chitin
C) Mucous membrane

Answer: B

2) What is the first line of defense in innate immunity?
A) Adaptive immune response
B) Barrier defenses
C) Antimicrobial peptides

Answer: B

3) What is the role of hemocytes in insect immunity?

A) They produce antibodies
B) They secrete antimicrobial peptides
C) They activate T cells

Answer: B

4) Which cells are involved in the inflammatory response by releasing histamine?

A) Neutrophils
B) Macrophages
C) Mast cells

Answer: C

5) What is the function of Toll-like receptors (TLRs) in mammals?

A) To produce antibodies
B) To recognize pathogen-associated molecular patterns
C) To secrete cytokines

Answer: B

6) What triggers the production of interferons in mammals?

A) Viral infections
B) Fungal infections
C) Bacterial infections

Answer: A

7) What is the main function of natural killer cells?

A) To detect and destroy abnormal cells
B) To activate B cells
C) To produce antibodies

Answer: A

8) What is the role of the complement system in immunity?

A) To lyse invading cells
B) To activate T cells
C) To produce antibodies

Answer: A

9) What is the primary function of antibodies in the immune response?

A) To kill pathogens directly
B) To activate T cells

C) To mark pathogens for destruction

Answer: C

10) What is the purpose of immunological memory?

A) To provide immediate immune response
B) To enhance the primary immune response
C) To provide long-term protection against diseases

Answer: C

11) What is the role of helper T cells in adaptive immunity?

A) To produce antibodies
B) To activate both humoral and cell-mediated immune responses
C) To destroy infected host cells

Answer: B

12) Which enzyme in insects breaks down bacterial cell walls?

A) Lysozyme
B) Amylase
C) Protease

Answer: A

13) What type of immune cells are responsible for phagocytosis in mammals?

A) B cells
B) T cells
C) Macrophages

Answer: C

14) What is the role of dendritic cells in the immune system?

A) To produce antibodies
B) To stimulate adaptive immunity
C) To secrete histamine

Answer: B

15) Which cells are primarily involved in the adaptive immune response?

A) Red blood cells
B) Lymphocytes
C) Platelets

Answer: B

16) What type of immunity is active immediately upon infection in all animals?

A) Innate immunity
B) Passive immunity
C) Adaptive immunity

Answer: A

17) Which cells secrete cytokines to recruit neutrophils during inflammation?

A) Mast cells
B) B cells
C) Natural killer cells

Answer: A

18) What is the function of antimicrobial peptides in the immune system?

A) To bind to antigens
B) To produce antibodies
C) To inactivate or kill pathogens

Answer: C

19) What triggers the inflammatory response in the body?

A) Antigen presentation
B) Antibody production
C) Molecules released upon injury or infection

Answer: C

20) Which immune cells are involved in the detection and destruction of cancerous cells?

A) B cells
B) Natural killer cells
C) Macrophages

Answer: B

21) What is the role of lysozyme in the immune system of insects?

A) To digest food
B) To produce antibodies
C) To break down bacterial cell walls

Answer: C

22) Which type of immune cell is responsible for secreting histamine during an allergic reaction?

A) Mast cells
B) Macrophages
C) Neutrophils

Answer: A

23) What is the primary function of neutrophils in immune system?

A) To produce antibodies
B) To engulf and destroy pathogens
C) To activate T cells

Answer: B

24) What is the role of interferons in the immune response?

A) To lyse bacterial cells
B) To activate B cells
C) To inhibit viral replication

Answer: C

25) Which component of the immune system is responsible for the lysis of invading cells?

A) Antibodies
B) Complement system
C) Cytokines

Answer: B

26) What is the main function of eosinophils in the immune system?

A) To attack parasites
B) To produce antibodies
C) To secrete histamine

Answer: A

27) What is the primary function of antibodies in the immune system?

A) To activate T cells
B) To kill pathogens directly
C) To mark pathogens for destruction

Answer: C

28) What is the purpose of immunological memory in the immune system?

A) To provide immediate response to pathogens
B) To enhance the primary immune response
C) To provide long-term protection against diseases

Answer: C

29) What is the role of helper T cells in the immune system?

A) To activate both humoral and cell-mediated immune responses
B) To produce antibodies
C) To destroy infected host cells

Answer: A

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True or false questions:

1) Insects do not have any defenses against viruses. (F)

2) Vertebrates have both innate and adaptive immunity. (T)

3) The inflammatory response is a feature of

adaptive immunity. (F)

4) Antibodies directly kill pathogens. (F)

5) Helper T cells are involved in both humoral and cell-mediated immune responses. (T)

6) Adaptive immunity is present in all animals, including invertebrates. (F)

7) Insects have an exoskeleton made of chitin that serves as a barrier to pathogens. (T)

8) Vertebrates have natural killer cells as part of their innate immune defenses. (T)

9) Antibodies are produced by T cells in response to antigens. (F)

10) The complement system is involved in both innate and adaptive immunity. (T)

11) MHC molecules are involved in the presentation of antigens to T cells. (T)

12) Monoclonal antibodies are produced from multiple clones of B cells. (F)

13) All animals have innate immunity, a defense active immediately upon infection. (T)

GOOD LUCK ...